

Homework 8

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This homework will be **extra credit**.

Problem 1

```
1  #include <reg51.h>
2
3  typedef enum { false, true } bool;
4  unsigned char R0;
5
6  void main() {
7      TMOD = 0x06; // Counter 2, Mode 2
8      TH0 = -0x60; // Initial value given
9
10     R0 = 0; // Just in case
11     TR0 = true; // Start the count
12
13     do {
14         if (TF0) {
15             R0++;
16             TF0 = false; // continue the count
17         }
18     } while (R0 != 60);
19
20     TR0 = false; // Stop the count
21 }
```

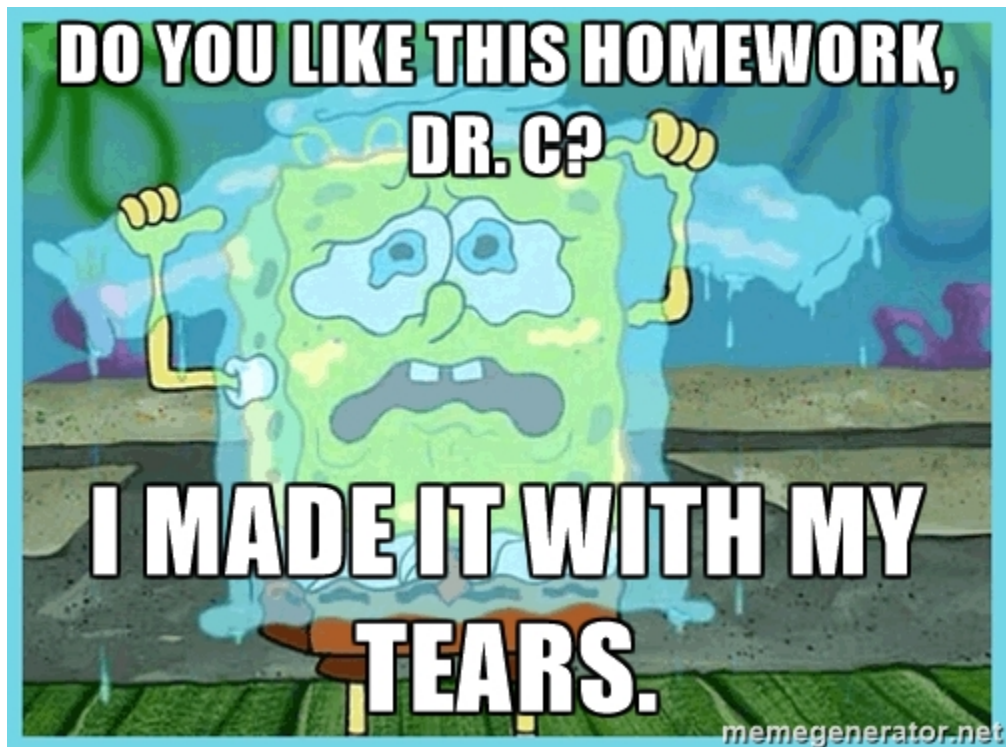
Problem 2

```
1      ORG 0H
2
3      LJMP MAIN
4
5      /* Timer 1 Interrupt */
6      ORG 001BH
```

```

7      CPL P1.1
8      RETI
9
10     /* Main */
11     ORG 0030H
12 MAIN: MOV P2, _num_00H      ; P2 = output port
13      MOV TMOD, _num_20H    ; Timer 1, mode 2
14      MOV TH1, #-37D        ; 5 kHz => 25000 Hz => 4^-5 s
15                               ; => 40 us => 40/1.085 ticks => 37 ticks
16      MOV IE, _num_88H
17      SETB TR1
18
19 LOOP: INC R0
20      MOV P2, R0            ; P2 = R0
21      ACALL DELAY
22      SJMP LOOP
23
24 ; so we need to generate a 10.85us delay
25 ; 10.85/1.085 = 10MC. Easier to just do a brute force
26 DELAY: MOV R1, _num_4D     ; 2 MC
27 HERE:  DJNZ R1, HERE        ; 4 * 2 = 8 MC
28      RET                   ; 8 + 2 = 10 MC
29
30
31     END

```

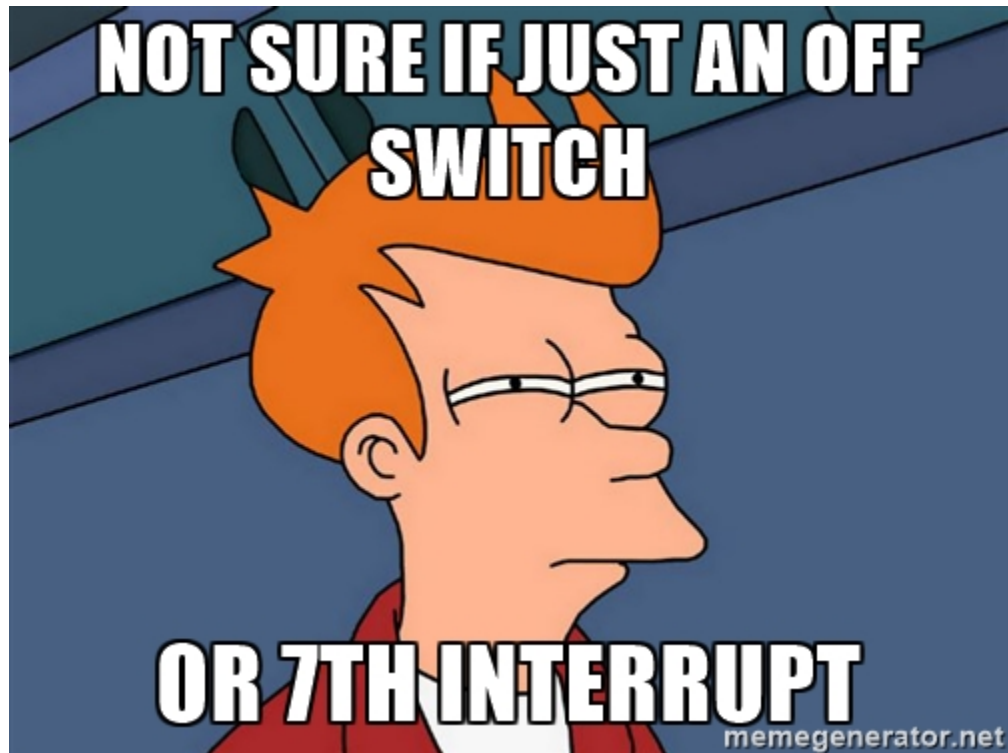


Problem 3

```
1      ORG 0
2      LJMP MAIN
3
4      /* Timer 0 Interrupt */
5      ORG 000B
6      CPL P1.1
7      RETI
8
9      /* Timer 1 Interrupt */
10     ORG 001B
11     CPL P1.2
12     RETI
13
14     /* Main */
15     ORG 30H
16 MAIN: MOV TMOD, _num_12H      ; T0M2 & T1M1
17     MOV P1, _num_0           ; P1 = output port
18     ACALL WAVE0
19     ACALL WAVE1
20
21     MOV IE, _num_8AH
22 REPEAT: SJMP REPEAT
23
24 ; 5kHz => 1/5000 s = .0002 s => 200 microseconds
25 ; 200 us => 184 ticks PER PERIOD. Assuming a 50%
26 ; duty cycle. So actual value is 184/2 = 92 ticks
27 WAVE0: MOV TH0, #-92
28     SETB TR0
29     RET
30
31 ; 500 Hz => .002 s => 2000 us
32 ; (FFFF - x + 1)(1.085) = 2000
33 ; x = FFFF - 1842 => E7 BD
34 WAVE1: MOV TH1, _num_0E7H
35     MOV TL1, _num_0BDH
36     SETB TR1
37     RET
38
39
40     END
```

Problem 4

```
1      ORG 0H
2      LJMP MAIN
3
4      /* Interrupts */
```



```
5      ORG 001BH
6      LJMP SWAVE
7      RETI
8
9      /* Main */
10     ORG 30H
11 MAIN:  MOV R7, _num_1D
12         MOV P1, _num_0           ; Make P1 an output port
13         MOV IE, _num_88H        ; Enable interrupt on T1
14
15 LOOP:  MOV P2, R7                ; R7 holds the value of the div series
16         ACALL INFSER
17         SJMP LOOP
18
19 ; code for the square wave interrupt
20 ; determines via R6 what the last wave was
21 SWAVE:  MOV TMOD, _num_20H      ; Timer 2, mode 2
22
23         CJNE R6, _num_70, SKIP2LOW
24         ACALL HWAVE
25         SJMP NEXT
26
27 SKIP2LOW:
28         ACALL LWAVE
29
30 NEXT:  SETB TR1
```

```

31         RET
32
33 ; the wave has a duty cycle of 70%
34 ; so the high wave needs a high wave of 7kHz
35 ; & kHz => 7000 Hz => 1.42^-4 s => 142 us => 130 ticks
36 ; 130.8 *.7 = 92 ticks for 70% duty cycle of a 7kHz wave
37 HWAVE:
38     SETB P1.1
39     MOV TH1, #-92
40     MOV R6, _num_70                ; Just to keep track of what our last
        value was
41     RET
42
43 ; so, exact same reasoning as before, we need 130 ticks for
44 ; the period of the wave, but now we just subtract 92 from 130
45 ; so, 130 - 92 = 38
46 LWAVE:
47     CLR P1.1
48     MOV TH1, #-38
49     MOV R6, _num_30
50     RET
51
52 ; computes infinite divergent series incrementally
53 ; uses R7
54 INFSER:
55     MOV A, R7
56     ACALL TWOCOMP
57     ACALL ISNEGATIVE
58
59     JC RETURN
60     INC A
61
62 RETURN: MOV R7, A
63     RET
64
65
66 ; returns two's compliment of A
67 TWOCOMP:
68     CPL A
69     ADD A, _num_1D
70     RET
71
72
73 ; returns 1 in C if B is negative
74 ISNEGATIVE:
75     CJNE A, _num_10000000B, NEXT2
76 NEXT2:  CPL C
77     RET
78
79
80     END

```

